
Early-Exit Speculative Decoding for Hindi Accelerating Large Language Models without Secondary Draft Models

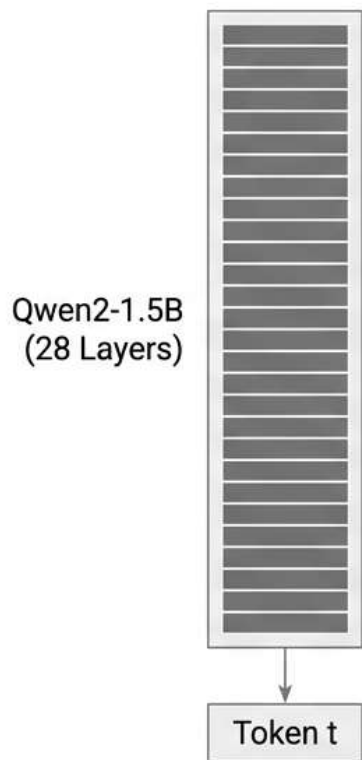
Team Transformers

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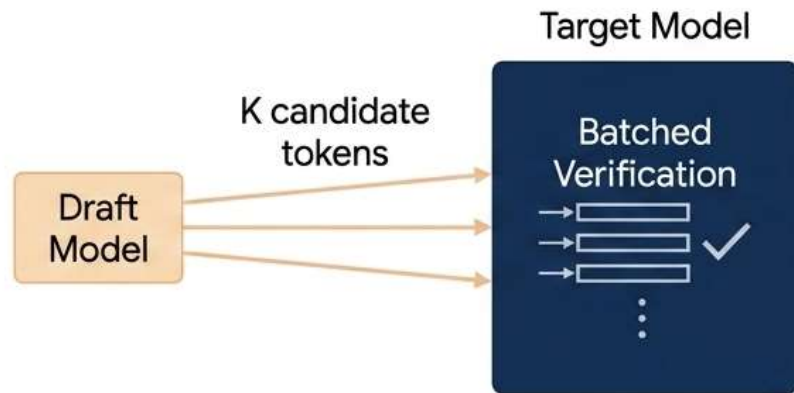
Sequential Generation is the Enemy of Speed



$$28 \text{ layers} \times 100 \text{ tokens} = 2,800 \text{ layer evaluations}$$

- Standard autoregressive (AR) generation produces exactly one token at a time: $x_t \sim p_{\theta}(x_t | x_{<t})$
- To generate a single word, the model must run a full forward pass through all 28 layers.
- Inference time scales linearly with both sequence length and model depth.

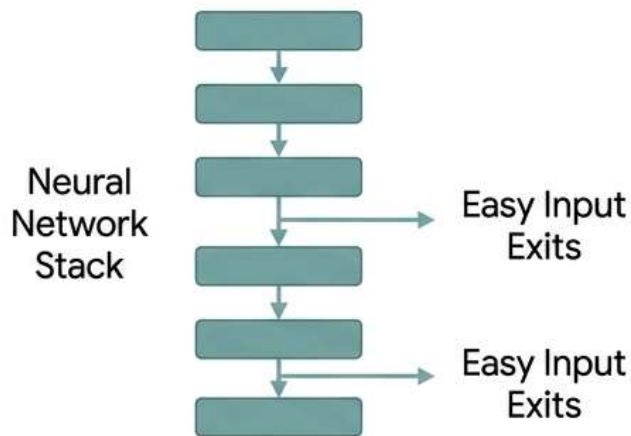
Idea 1: Speculative Decoding



Mechanism:	A fast draft model proposes K candidate tokens. The large target model verifies all K tokens in a single batched forward pass.
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Drawback:	Requires maintaining, syncing, and storing two separate models.
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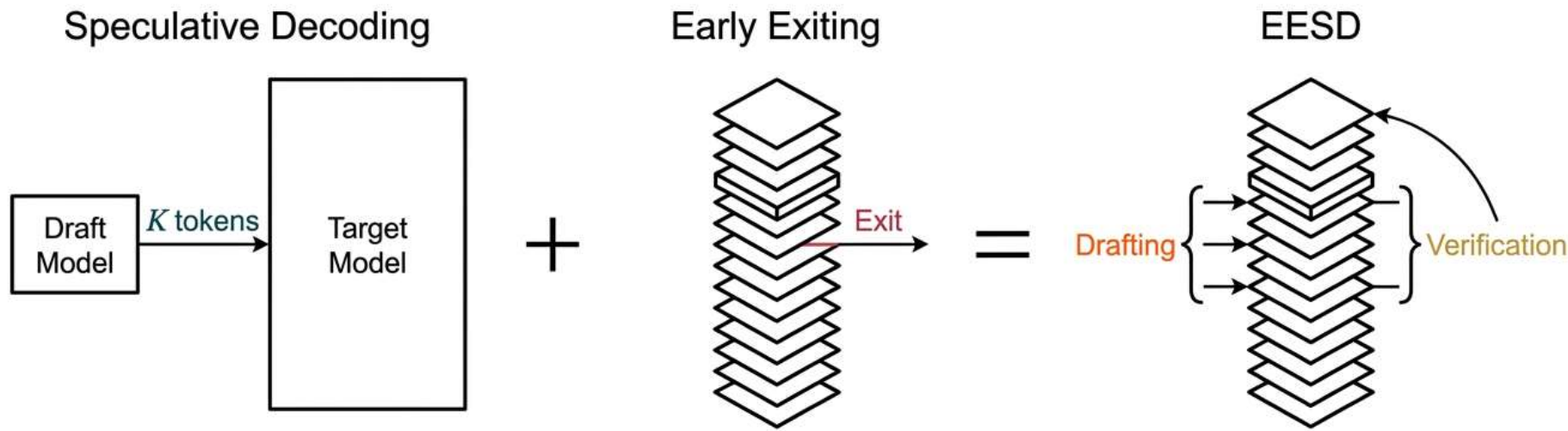
Idea 2: Early Exiting



Mechanism:	Attach classifiers at intermediate layers. Route easy inputs to shallow exits to save compute.
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Drawback:	Still generates tokens one by one; does not actually solve the sequential bottleneck.
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The Synthesis: Speculative Decoding Meets Early Exiting

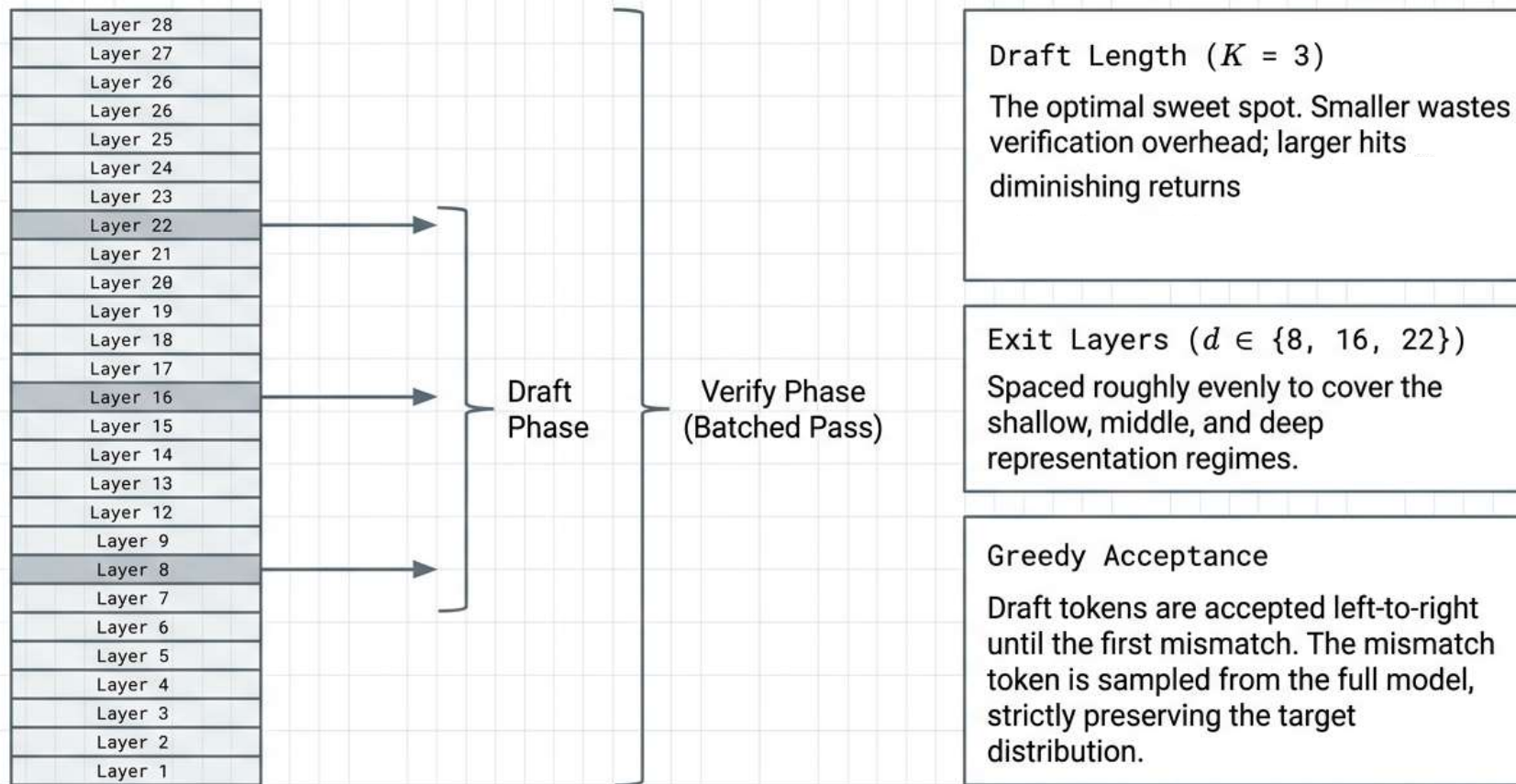


Mechanism: Fast model guesses K tokens; big model verifies in one batched pass.
Flaw: Requires hosting two separate models.

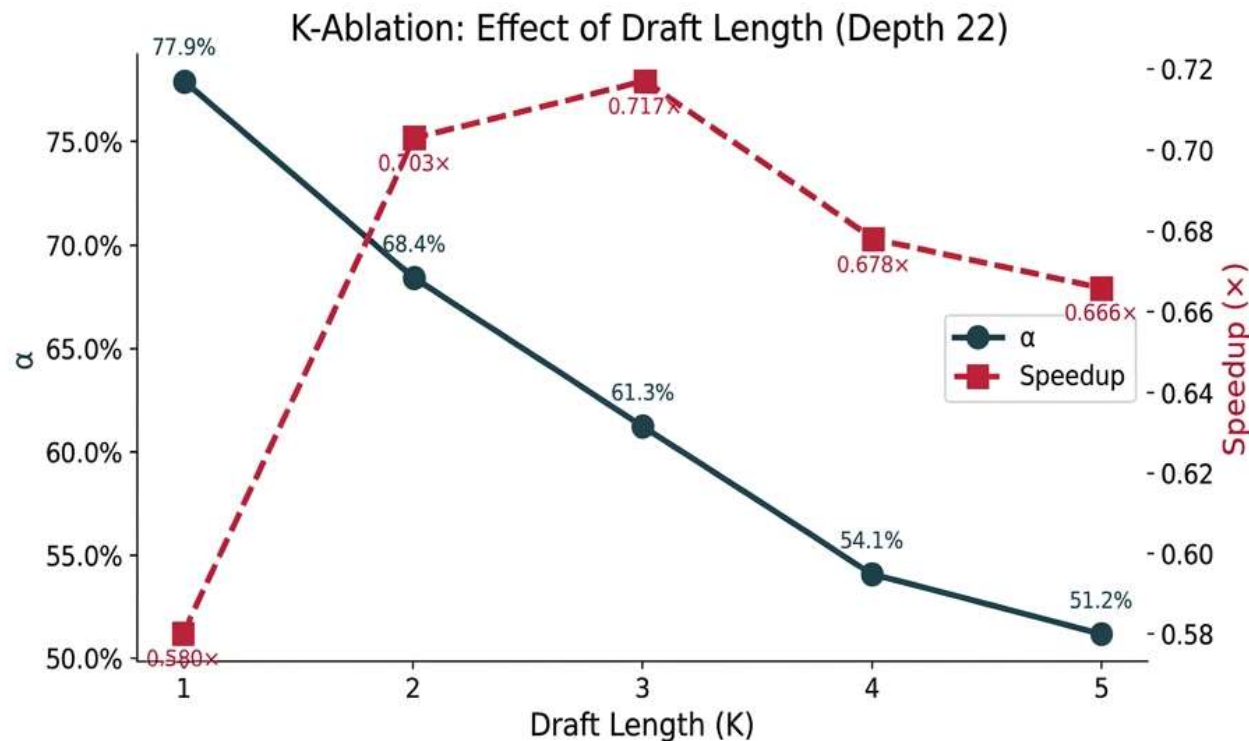
Mechanism: Stopping computation mid-network.
Flaw: Still generates one token at a time.

Mechanism: Use early layers of the same model to draft, and the full model to verify.
One model, multiple exits.

The Foundational Mechanics of Early-Exit Speculative Decoding



Hyperparameter Discipline: The K-Ablation Sweet Spot



Ablation Rationale:

Changing one variable (K) while fixing depth (22) and policy isolates the exact effect of draft length.

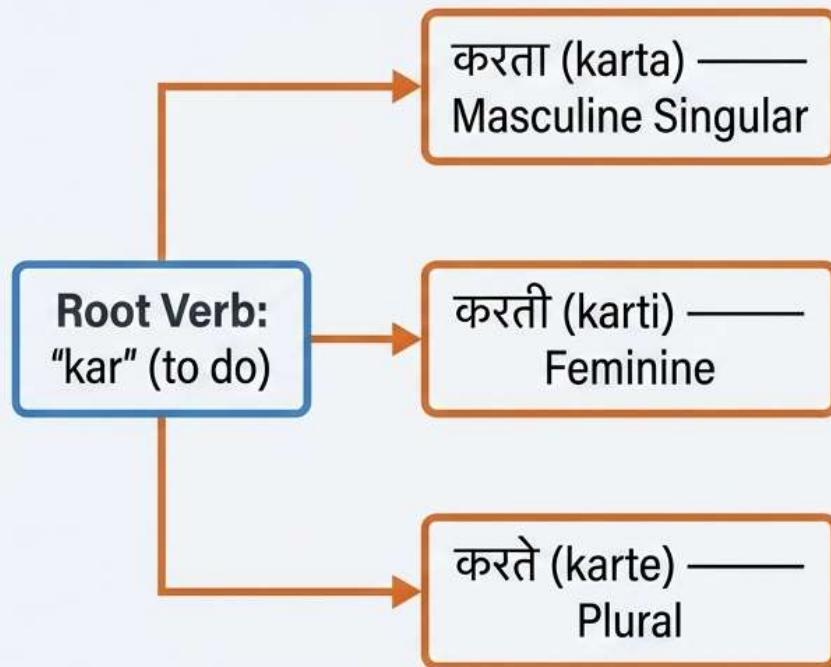
The Finding: Why $K = 3$?

- At $K = 1, 2$: Verification overhead outpaces drafting gains.
- At $K > 3$: Acceptance drops sharply as compounding errors multiply.
- $K = 3$ represents the absolute mathematical apex for wall-clock speedup.

Why Hindi? Rigorous Inflectional Friction

High-Entropy Environment

Hindi possesses an exceptionally rich morphological structure featuring inflected verbs, complex postpositions, and multi-axis agreement.

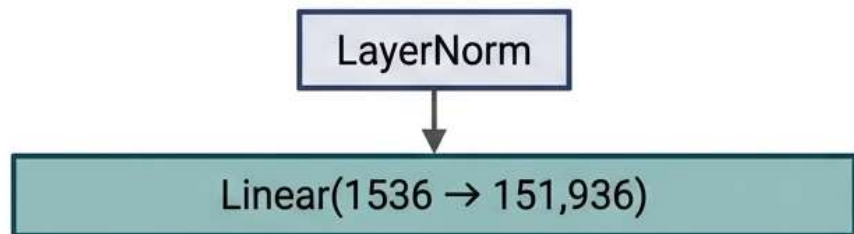


Stress-Testing the Architecture

Base forms change form heavily depending on gender, number, and context. This forces the early-exit heads to capture deep syntactical context rather than relying on shallow memorization.

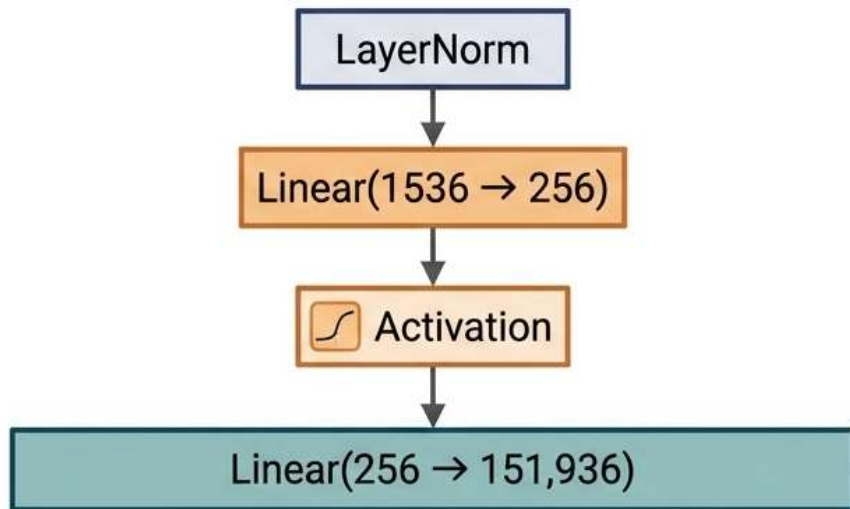
Placement: Exit heads attached at three strategically spaced depths: Layer 8 (Shallow), Layer 16 (Middle), Layer 22 (Deep).

Architecture 1: Hook Head



The simplest baseline; hooks directly into the hidden state.

Architecture 2: Bottleneck Head (BN)



The Advantage: Dimensionality reduction acts as a powerful regularizer, forcing compressed, essential representations over mere memorization.

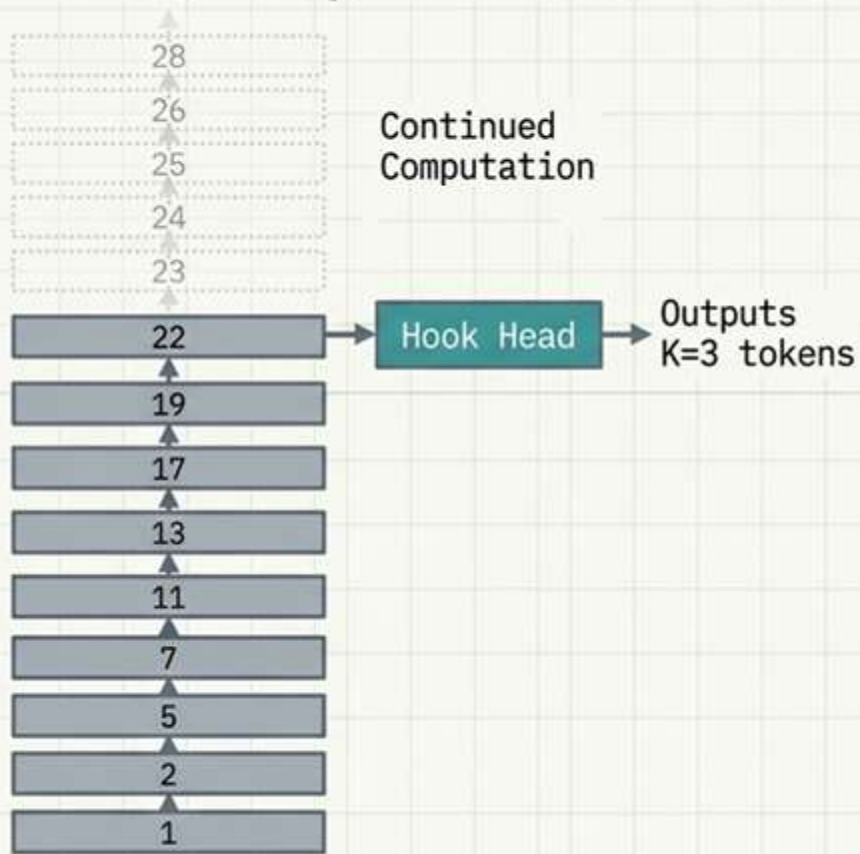
Experimental Taxonomy of 12 Decoding Strategies

Training Protocol: Exit heads trained via Teacher Forcing and Knowledge Distillation (Temperature $T=2$).

Baselines	Autoregressive (AR) Draft Model
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	Every policy tested across both head architectures for clean variable ablation.	
	Hook (Standard Head)	Bottleneck (Compressed Head)
Fixed-Depth Policies	TrueExit	BN-TrueExit
Adaptive Bandits	Thompson Sampling UCB1 ($c = \sqrt{2}$)	Thomp-BN UCB-BN
Reward Shaped	Entropy Weighted Thompson	WThomp-BN (0.92/0.08 reward split)

Fixed-Depth Baseline: The Hook Architecture



Mechanism

Always drafts from the deepest available exit ($d=22$) because it yields the highest raw acceptance rate (α).

Architecture

LayerNorm $\rightarrow$ Linear(1536 $\rightarrow$ 151,936)

The simplest possible design. It "hooks" the hidden state but does not alter it.

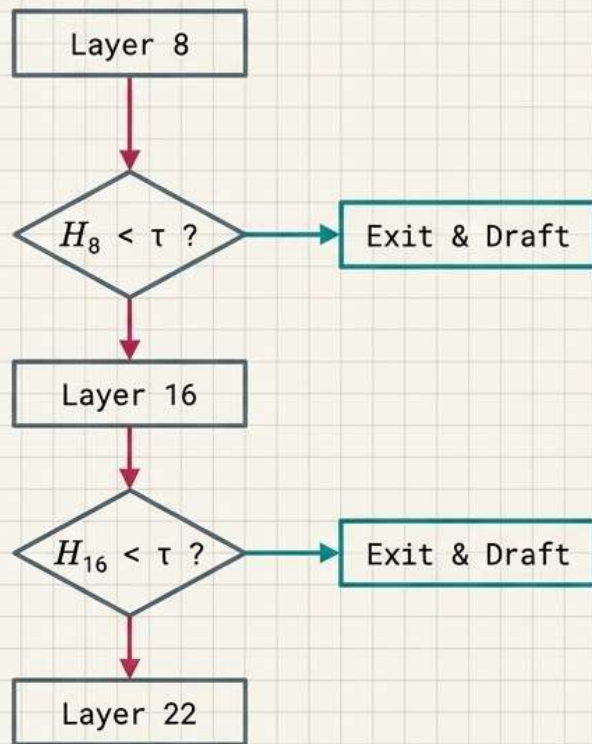
Design Justification

Included strictly as an A/B Control. Any performance gap between Hook and TrueExit is due entirely to stopping the forward pass, not head capability.

Deterministic Gating: The Entropy Strategy

Shannon Entropy:

$$H_d = -\sum p_v \log p_v$$



Mechanism

Pick the shallowest layer whose output entropy falls below threshold τ . Low entropy implies high confidence.

The Diagnostic Failure

Entropy massively underperforms (0.57x speedup). Why? Because confidence is not correctness. An exit head can be confidently wrong (low entropy, incorrect token) or uncertainly right (high entropy, correct token). Furthermore, a single threshold τ does not cleanly generalize across layers.

Directed Exploration: Upper Confidence Bound (UCB)

$$\text{UCB}_d = \underbrace{\left[\hat{\mu}_d \right]}_{\text{Empirical Mean}} + \underbrace{\left[c \sqrt{\frac{\ln N}{n_d}} \right]}_{\text{Exploration Bonus}} \quad \leftarrow \quad \begin{array}{l} c = \sqrt{2} \approx 1.41 \\ \text{(Textbook Hoeffding} \\ \text{bound constant)} \end{array}$$

Mechanism: A deterministic multi-armed bandit.

Explicitly favors the arm with the highest established reward but forces occasional exploration of neglected arms via the logarithmic bonus.

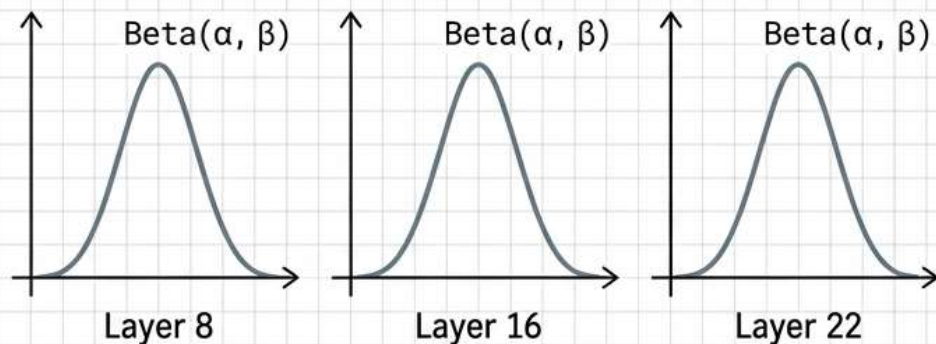
Comparison: UCB-BN significantly outperforms UCB-Hook (0.919x vs 0.578x), proving that bottleneck architecture yields superior baseline rewards per arm.

The Reward Shaping Necessity

Unlike Thompson, UCB is deterministic and over-commits quickly. If rewarded solely on correctness, it permanently locks onto Layer 22.

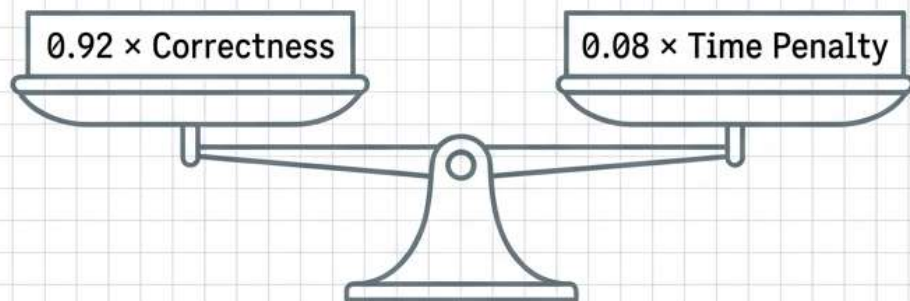
To force UCB to utilize shallow layers, it absolutely requires the latency-weighted reward structure (0.92 correctness + 0.08 time).

Adaptive Intelligence: The Thompson Sampling Family



Mechanism: A Bayesian multi-armed bandit. Start with a uniform prior $\text{Beta}(1, 1)$. After every draft, observe accepted/rejected tokens and update beliefs. Exploration naturally decays.

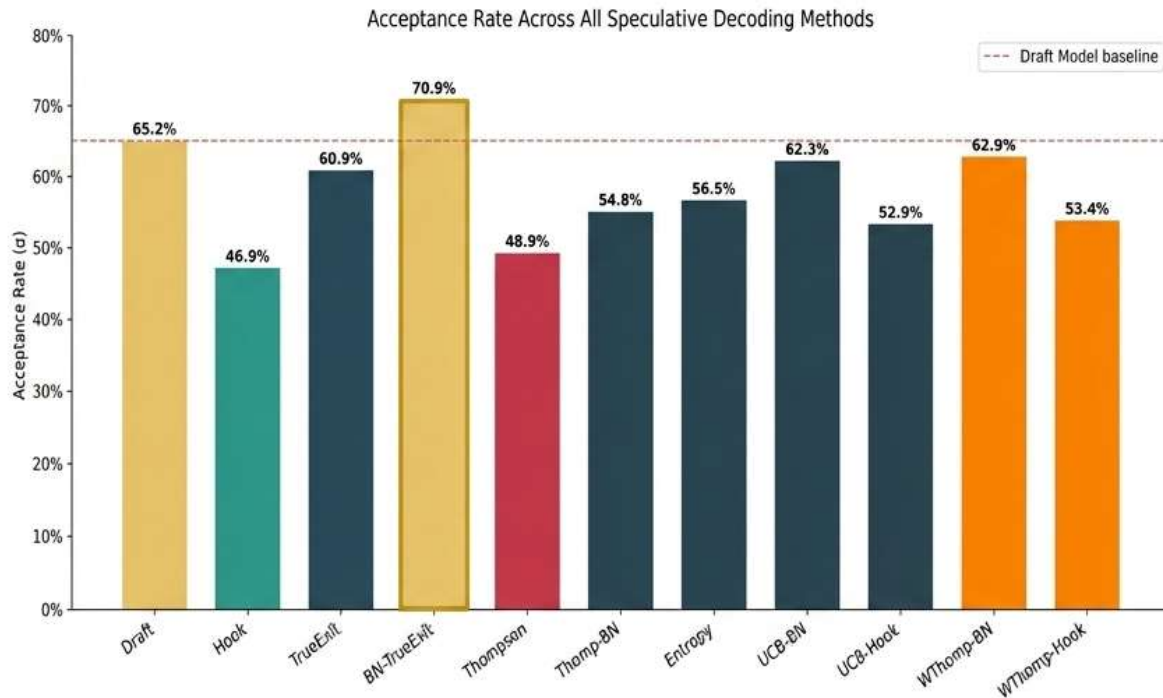
Result: Thomp-BN is the overall champion, yielding the highest wall-clock speedup (1.130x).



Design Insight: Weighted Variants (WThomp)

Why did weighting hurt Thompson's speedup? Thompson's stochastic nature naturally explores "cheap" shallow layers. Explicitly rewarding time biases the policy too far away from the highly accurate accurate Layer 22, breaking the optimal balance.

BN-TrueExit Dominates Distribution Fidelity

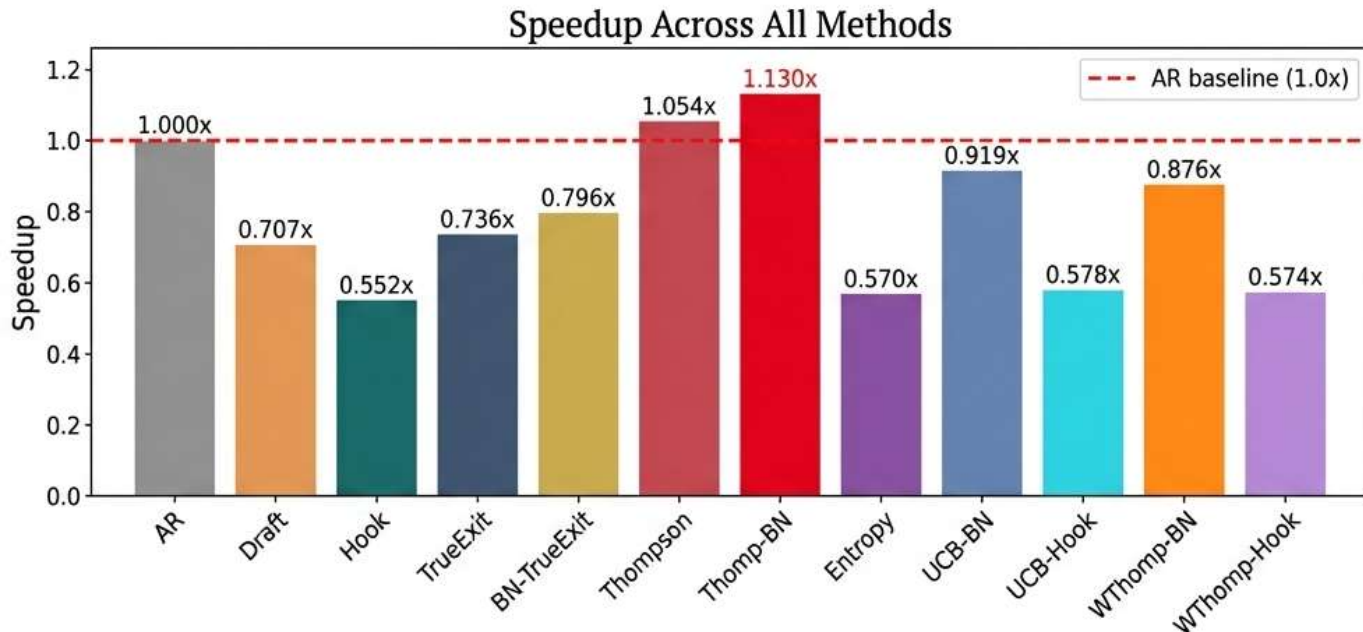


- BN-TrueExit achieves a 70.9% acceptance rate (α).
- Nearly 3 out of 4 draft tokens exactly match the full model's distribution.
- Standard Draft sits at 65.2%.
- Adaptive methods (like Thompson and Thomp-BN) sit in the mid-50% range because they frequently sample from shallower, less accurate layers.

The Speedup Reality: High Acceptance is Insufficient

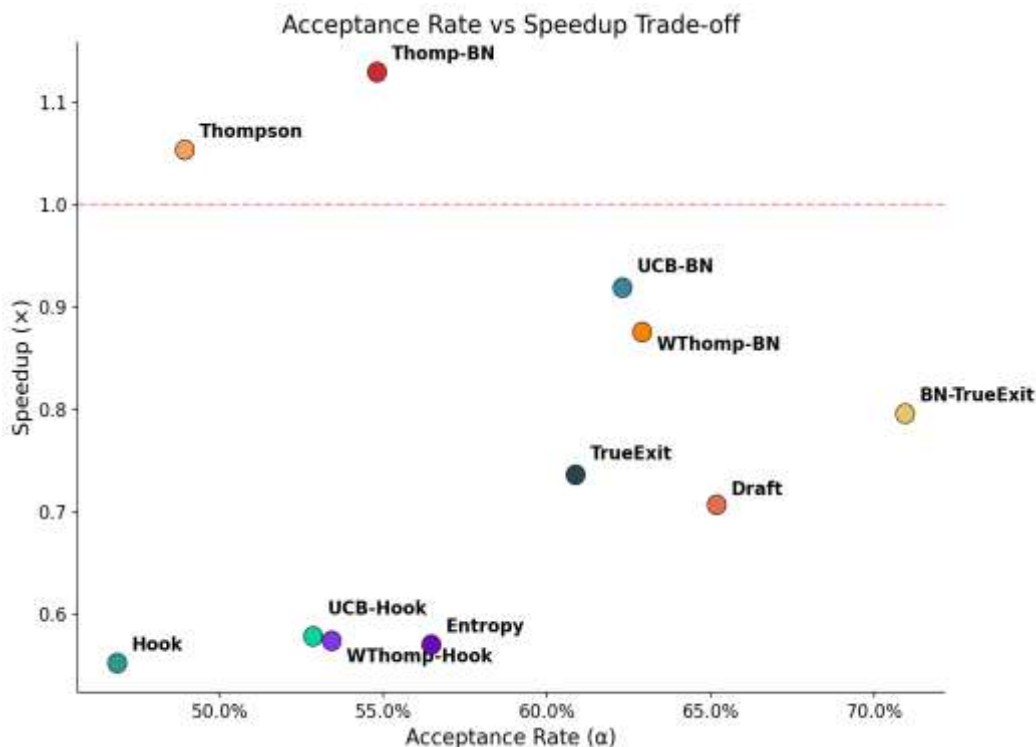
Previous Assumption

- BN-TrueExit
Acceptance Rate:
70.9%
- Highest α in the grid.



- BN-TrueExit drops below baseline speed (0.796x) due to massive drafting overhead.
- Only adaptive methods cross the 1.0x wall-clock barrier.
- Thomp-BN achieves 1.130x speedup and a throughput of 30.40 Tokens/sec (a 14.4% gain over AR baseline).

The Performance Landscape: Three Distinct Trade-Off Clusters



Cluster 1: High Fidelity, Moderate Speed (Top Right)

- BN-TrueExit, UCB-BN
- Prioritizes matching the exact distribution over latency.

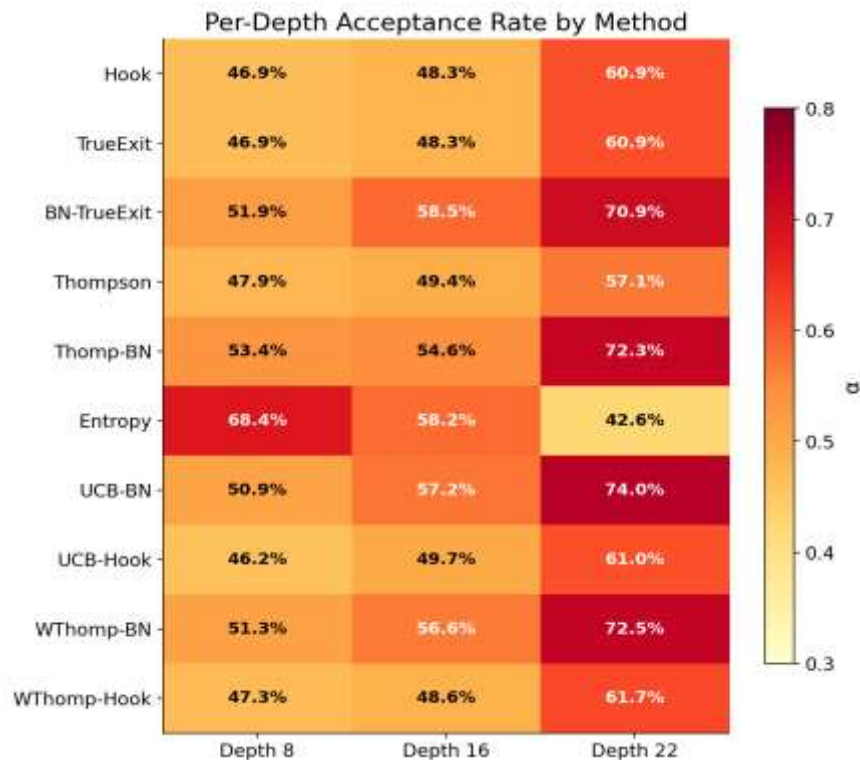
Cluster 2: High Speed, Moderate Fidelity (Top Left)

- Thompson-BN, Thompson
- Exploits adaptive depth selection for maximum wall-clock gains.

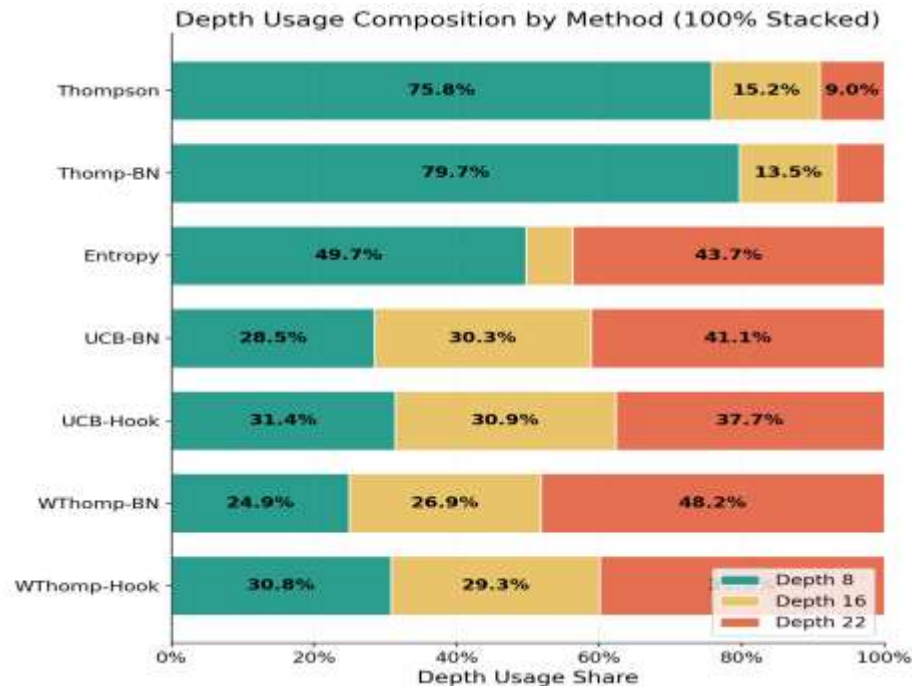
Cluster 3: Low Efficiency (Bottom Left)

- Hook-based methods

Mechanistic Analysis: Why Adaptive Depth Outperforms



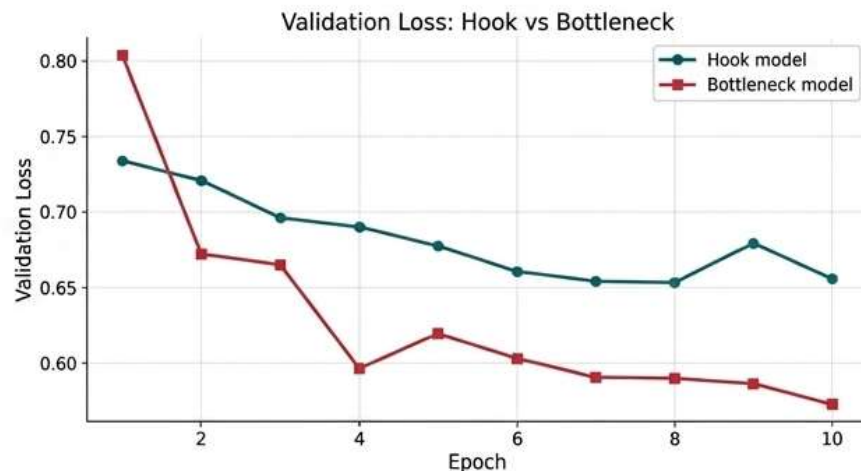
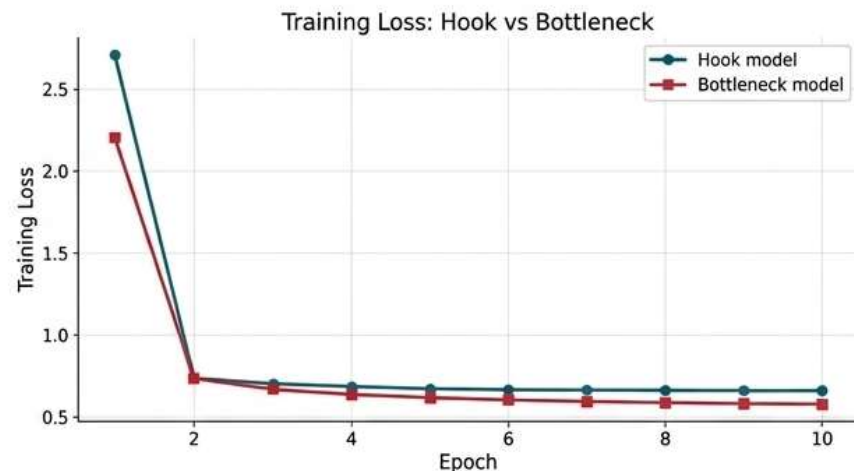
A strict monotonic rule. Layer 22 consistently yields higher accuracy than 16 or 8.



Thompson-BN does not default to Layer 22. It selects the shallowest, cheapest exit (Layer 8) almost 80% of the time, keeping drafting costs minimal when predictions are good enough.

Training Diagnostics: Compression Acts as Regularization

Bottleneck heads strictly outperform standard Hook heads across every policy and metric. The training diagnostics explain why.



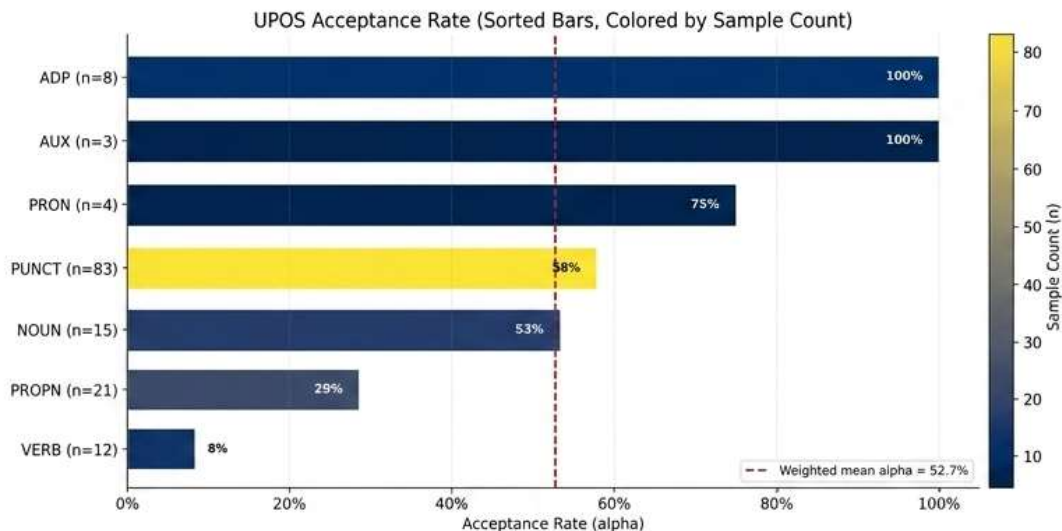
The Mechanism

The dimensional bottleneck (compressing the hidden state to 256) forces the network to retain only the highest-signal features.

The Result

This structural constraint acts as a powerful regularizer, preventing overfitting. Validation loss tracks training loss perfectly, yielding genuine generalization.

The Linguistic Bottleneck: UPOS Morphological Analysis



Methodology

Fixed-depth 22 ablation used to isolate linguistic difficulty from adaptive policy shifts.

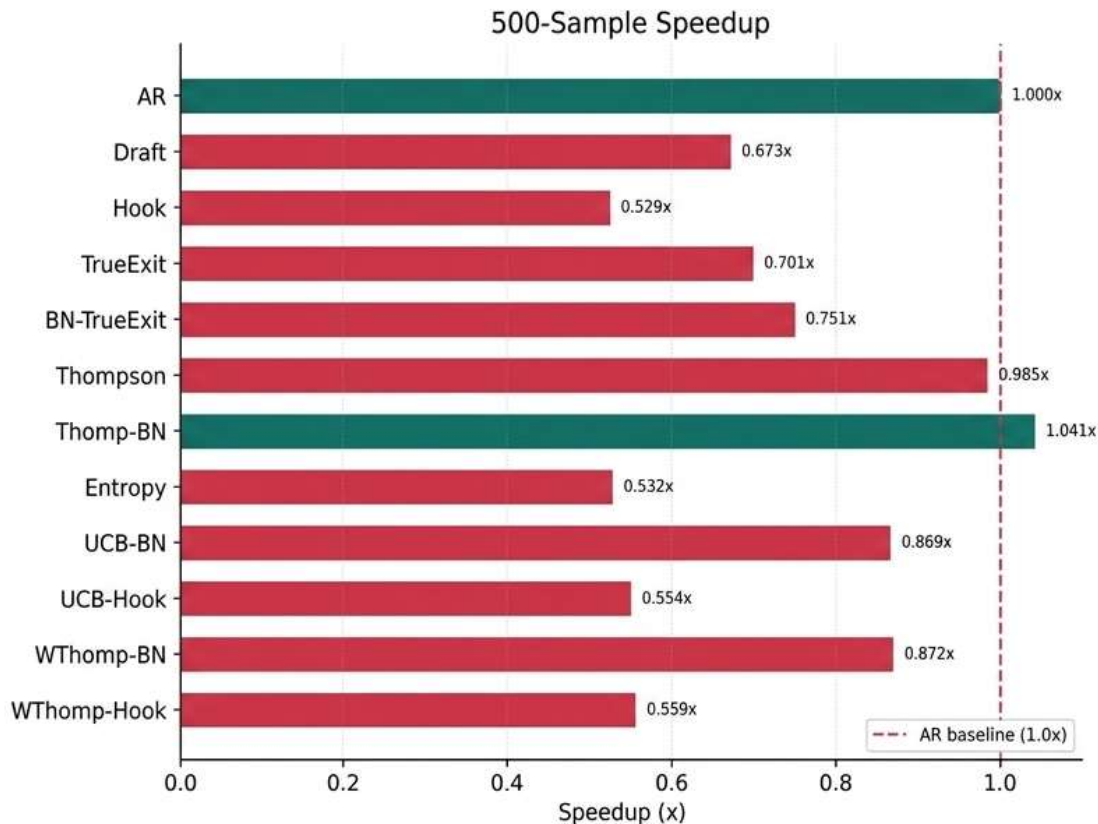
Closed-Class (Predictable)

Postpositions (ADP) and Auxiliaries (AUX) achieve 100% acceptance. Governed strictly by syntactic context.

Open-Class (Complex)

Hindi Verbs plummet to 8%. Verbs agree in person, number, and gender (e.g., करता / करती / करते). Early exit representations lack sufficient global subject context to predict the exact inflection.

Evaluating Robustness at Scale (500 OOD Prompts)



The Academic Stress Test:

Scaling from 101 samples to 500 out-of-distribution (OOD) Hindi prompts.

Results Sustained:

- Rankings are perfectly preserved. BN-TrueExit remains the α leader (68.8%).
- Thomp-BN decisively sustains $> 1.0x$ wall-clock speedup (1.041x).

Conclusion:

The adaptive bottleneck advantage is a structural reality, not a small-sample artifact.

Deployment Matrix: Engineering Recommendations

Latency-Critical Systems	Quality/Fidelity-Critical Systems	Balanced Workloads
Recommendation: Thomp-BN Profile: Peak speed (1.13x) and maximum throughput (30.40 Tok/s). Best when raw generation time is the strict priority.	Recommendation: BN-TrueExit Profile: Peak acceptance (70.9%). Best when generation fidelity cannot be compromised, even at the cost of wall-clock speed.	Recommendation: UCB-BN Profile: A mathematically bounded compromise. Maintains high acceptance (~60%) with minimal latency penalties (0.92x).

Inferences

1. **Adaptive > Fixed:** Theoretical acceptance (α) is insufficient. Adaptive depth selection is mandatory for real-world speedups.
2. **Bottleneck > Hook:** Dimensional compression acts as a crucial regularizer, universally improving both training stability and inference speed.
3. **The Morphological Penalty:** EESD struggles specifically with highly inflected word classes (verbs), directly mapping computational latency to linguistic structure.

THANK YOU